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Big Data Analytics | Assignment 1: Analysis of SVD for Fine-Tuning LLMs (GitHub)

Question 1.

Accuracy	Original LLM	Fine-Tuned LLM (RANK = 16)	Fine-Tuned LLM (RANK = 64)	Fine-Tuned LLM (RANK = 256)
GSM8K	0.032	0.153	0.182	0.221
MATH	0.001	0.031	0.040	0.039

Fine-tuning with PiSSA improved the performance of Llama-3.2-1B on both GSM8K and MATH compared with the original pretrained model. On GSM8K, accuracy increased from 0.032 to 0.153, 0.182, and 0.221 for ranks 16, 64, and 256, respectively. On MATH, accuracy improved from 0.001 to 0.031, 0.040, and 0.039.

For GSM8K, performance consistently improved as the target rank increased. A higher rank includes more singular components in the PiSSA initialization and increases the number of trainable low-rank parameters. Therefore, the model has a larger adaptation capacity, or more degrees of freedom to learn task-specific parameter updates. Rank 256 consequently achieved the best GSM8K accuracy.

For MATH, the improvement was not strictly monotonic. Rank 64 achieved 0.040, slightly higher than 0.039 at rank 256. This indicates that increasing the rank does not always guarantee better generalization. Performance is also affected by optimization behavior, task difficulty, and the usefulness of the additional trainable dimensions.

Question 2.

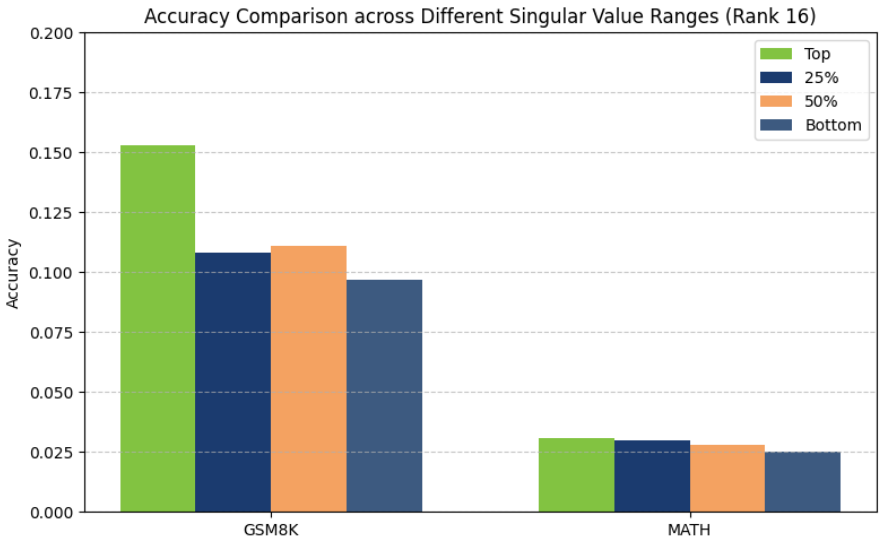
	Fine-Tuned LLM (RANK = 16)	Fine-Tuned LLM (RANK = 64)	Fine-Tuned LLM (RANK = 256)
Training Time (min.)	23.011705	23.270002	27.083443

Training time increased as the target rank increased. Rank 16 required 23.01 minutes, rank 64 required 23.27 minutes, and rank 256 required 27.08 minutes.

This occurs because a higher rank increases the size of the two trainable PiSSA low-rank matrices, matrix A and B. Consequently, more parameters must be processed during the forward pass, gradient computation, and parameter update. Therefore, rank 256 requires more computation than ranks 16 and 64. The results show a trade-off between adaptation capacity and computational efficiency: a larger rank may improve accuracy, but it also increases training cost.

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Question 3.



Accuracy	Top ranges	Quarter (25%) ranges	Middle (50%) ranges	Bottom ranges
GMS8K	0.153	0.108	0.111	0.097
MATH	0.031	0.03	0.028	0.025

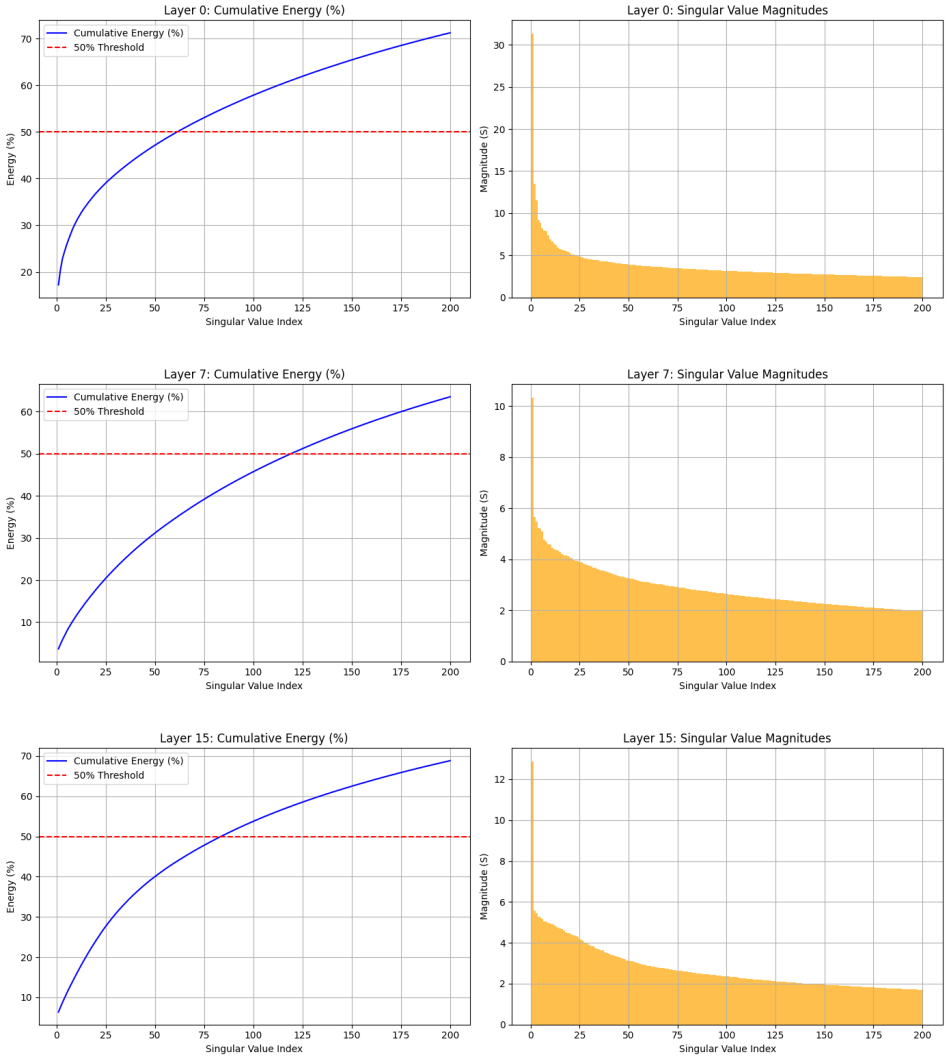
The Top-16 singular components achieved the highest accuracy on both datasets. On GSM8K, the accuracies for the Top, 25%, 50%, and Bottom ranges were 0.153, 0.108, 0.111, and 0.097. On MATH, the corresponding results were 0.031, 0.030, 0.028, and 0.025.

This result is consistent with the SVD interpretation. Singular values are ordered from largest to smallest, and larger singular values correspond to components that contribute more strongly to the original weight matrix. Therefore, using the Top-16 components initializes PiSSA from the more dominant directions of the pretrained weights.

The GSM8K middle-range result of 0.111 is slightly higher than the 25%-range result of 0.108, but the difference is very small. Such variation can occur because downstream accuracy is not determined only by singular-value magnitude, but also by optimization and task-specific interactions. Overall, the results still show that top singular components provide the most effective initialization, while the bottom range gives the lowest performance.

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Question 4.



For the q_proj weights, the minimum ranks required to retain at least 50% cumulative energy were approximately 62 for Layer 0, 120 for Layer 7, and 84 for Layer 15.

This indicates that the singular-value energy is distributed differently across layers. Layer 0 reaches 50% energy using fewer components, which suggests that its spectrum is more concentrated in the leading singular directions. Layer 7 requires the largest number of components, indicating a more distributed spectrum, while Layer 15 lies between the two.

The cumulative-energy plots support this interpretation: a faster-rising curve indicates that fewer singular components capture a large proportion of the weight information, while a slower-rising curve indicates that the information is spread across more components. This analysis can be used as a reference for selecting PiSSA ranks, although the rank that preserves a certain amount of SVD energy does not necessarily guarantee the highest downstream accuracy.